Algebra II with Trigonometry Chapter 7.1

Study Guide (Gemini)

Saved tutoring response with MathJax rendering. Original PDF file: 4564659_alg-2trig-h-chp-7-1-note.pdf

Here is a comprehensive breakdown of the attached PDF on trigonometric identities, formatted as requested.

1. Short Summary

This document is a guided note sheet or worksheet for an Algebra 2/Trigonometry Honors class (Chapter 7.1). It introduces fundamental trigonometric identities—Quotient, Product, Reciprocal, Pythagorean, and Cofunction—and uses a "Magic Hexagon" as a visual tool to help students memorize them. The text outlines step-by-step strategies for simplifying trigonometric expressions and formally proving complex identities, followed by a wide variety of practice problems ranging from basic simplification to trigonometric substitution.

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2. Core Definitions, Theorems, and Formulas

Definitions:

- **Identity:** An equation that is true for all values of the variable(s).
- **Trigonometric Identity:** An identity involving trigonometric functions.

Formulas & Properties (from the "Magic Hexagon" and lists):

- **Quotient Properties:**
- $\tan(x) = \frac{\sin(x)}{\cos(x)}$
- $\cot(x) = \frac{\cos(x)}{\sin(x)}$ (implied via hexagon)
- **Product Properties:**

- $\tan(x)\cos(x) = \sin(x)$
- $\tan(x)\cot(x) = 1$
- **Reciprocal Properties:**
- $\sin(x) = \frac{1}{\csc(x)}$
- $\cos(x) = \frac{1}{\sec(x)}$
- $\tan(x) = \frac{1}{\cot(x)}$
- **Pythagorean Properties:**
- $\sin^2(x) + \cos^2(x) = 1$
- $\tan^2(x) + 1 = \sec^2(x)$
- $1 + \cot^2(x) = \csc^2(x)$
- **Cofunction Identities:**
- $\sin(\frac{\pi}{2} - x) = \cos(x)$
- $\tan(\frac{\pi}{2} - x) = \cot(x)$
- $\sec(\frac{\pi}{2} - x) = \csc(x)$
- $\cos(\frac{\pi}{2} - x) = \sin(x)$
- $\cot(\frac{\pi}{2} - x) = \tan(x)$
- $\csc(\frac{\pi}{2} - x) = \sec(x)$

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3. Worked Study Guide

Goal 1: Simplifying Trigonometric Expressions The aim is to use identities to rewrite complicated expressions into simpler ones.

- *Strategy:* Convert everything to sines and cosines, use common denominators, or factor.
- *Walkthrough (Example 1a):* Simplify $\sin\theta\sec\theta$.

1. Convert $\sec\theta$ using the Reciprocal Property: $\sec\theta = \frac{1}{\cos\theta}$. 2. Multiply: $\sin\theta \cdot \frac{1}{\cos\theta} = \frac{\sin\theta}{\cos\theta}$. 3. Apply Quotient Property: $\frac{\sin\theta}{\cos\theta} = \tan\theta$.

Goal 2: Proving Trigonometric Identities To prove an identity, you must show that one side of the equation can be transformed into the other using established rules.

- *Guidelines:* Start with the most complicated side. Use algebra (factoring, finding common denominators) and fundamental identities. Never move terms across the equals sign; treat each side independently!

- *Walkthrough (Example 3d):* Verify $\frac{\sec A-1}{\sec A+1} = \frac{1-\cos A}{1+\cos A}$

1. Start with the left side (it has reciprocal functions, which are easy to convert): $\frac{\sec A-1}{\sec A+1}$ 2. Convert to cosines: $\frac{\frac{1}{\cos A}-1}{\frac{1}{\cos A}+1}$ 3. Clear the complex fractions by multiplying the numerator and denominator by $\cos A$: $\frac{(\frac{1}{\cos A}-1)\cdot\cos A}{(\frac{1}{\cos A}+1)\cdot\cos A}$ 4. Distribute: $\frac{1-\cos A}{1+\cos A}$. This matches the right side, so the identity is verified.

Goal 3: Trigonometric Substitution This is a tool used in upper-level math (like Calculus) to remove square roots.

- *Walkthrough (Example 4a):* Simplify $\sqrt{x^2-1}$ by substituting $x = \sec\theta$ (assume $0 < \theta < \frac{\pi}{2}$).

1. Substitute: $\sqrt{(\sec\theta)^2-1}$ 2. Use the Pythagorean Identity ($\tan^2\theta + 1 = \sec^2\theta \rightarrow \sec^2\theta - 1 = \tan^2\theta$). 3. Replace the expression under the root: $\sqrt{\tan^2\theta}$ 4. Simplify: $\tan\theta$. (It is positive because the angle is in the first quadrant, $0 < \theta < \frac{\pi}{2}$).

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4. Practice Problems

These problems are taken directly from the examples in the PDF.

Problem 1 (From Ex 1b): Simplify the expression: $\tan^2 x - \sec^2 x$ **Answer 1:** By the Pythagorean property, $\tan^2 x + 1 = \sec^2 x$. Rearranging this equation by subtracting $\sec^2 x$ from both sides and subtracting 1 from both sides yields $\tan^2 x - \sec^2 x = -1$.

Problem 2 (From Ex 1c): Simplify the expression: $\sin^2\left(\frac{\pi}{2}-y\right)\sec y$ **Answer 2:** 1. Use the cofunction identity $\sin\left(\frac{\pi}{2}-y\right) = \cos y$. Because it is squared, we get $\cos^2 y$. 2. The expression becomes $\cos^2 y \sec y$. 3. Use the reciprocal identity $\sec y = \frac{1}{\cos y}$. 4. Multiply: $\cos^2 y \cdot \frac{1}{\cos y} = \cos y$.

Problem 3 (From Ex 2a): Simplify the expression: $\frac{\cos x \sec x}{\cot x}$ **Answer 3:** 1. Focus on the numerator. Using the reciprocal identity,

$\cos x \sec x = \cos x(\frac{1}{\cos x}) = 1$. 2. The expression becomes $\frac{1}{\cot x}$. 3. Using the reciprocal identity, $\frac{1}{\cot x} = \tan x$.

Problem 4 (From Ex 3a): Verify the identity: $\frac{\cot x \sec x}{\csc x} = 1$ **Answer 4:** 1. Start with the left side and convert everything to sines and cosines. 2. $\cot x = \frac{\cos x}{\sin x}$ and $\sec x = \frac{1}{\cos x}$ 3. Multiply the numerator parts: $\frac{\cos x}{\sin x} \cdot \frac{1}{\cos x} = \frac{1}{\sin x}$. 4. Recognize that $\frac{1}{\sin x} = \csc x$. 5. Substitute this back over the denominator: $\frac{\csc x}{\csc x} = 1$. The identity is verified.

Problem 5 (From Ex 4b): Make the indicated trigonometric substitution and simplify: $\frac{1}{x^2\sqrt{4+x^2}}$, let $x = 2\tan\theta$ (Assume $0 < \theta < \frac{\pi}{2}$). **Answer 5:** 1. Substitute $2\tan\theta$ for x: $\frac{1}{(2\tan\theta)^2\sqrt{4+(2\tan\theta)^2}}$ 2. Square the terms: $\frac{1}{4\tan^2\theta\sqrt{4+4\tan^2\theta}}$ 3. Factor out the 4 inside the radical: $\sqrt{4(1+\tan^2\theta)} = 2\sqrt{1+\tan^2\theta}$. 4. Use Pythagorean identity ($1+\tan^2\theta = \sec^2\theta$): $2\sqrt{\sec^2\theta} = 2\sec\theta$. 5. Put it all back together: $\frac{1}{4\tan^2\theta \cdot 2\sec\theta} = \frac{1}{8\tan^2\theta\sec\theta}$. 6. (Optional simplification to sines/cosines): $\frac{1}{8\cdot(\frac{\sin^2\theta}{\cos^2\theta})\cdot(\frac{1}{\cos\theta})} = \frac{\cos^3\theta}{8\sin^2\theta}$.

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5. Assumptions and Ambiguities Resolved

- **Blank Even-Odd Identities:** On Page 2, the document says, "List even-odd identities for trigonometric functions," but leaves the space completely blank. However, Example 2b and Example 3c specifically require these identities to solve (e.g., needing to know that $\cos(-x) = \cos x$ and $\sin(-x) = -\sin x$). I assumed the standard even-odd identities apply and would have been provided verbally by the teacher.
- **"Magic Hexagon" Extrapolation:** The PDF gives explicit formulas for only the clockwise quotient ($\tan = \frac{\sin}{\cos}$) and one product ($\tan \cdot \cos = \sin$). It implies, but does not explicitly write out, the rest of the hexagon's logic (e.g., going counter-clockwise $\cot = \frac{\cos}{\sin}$). I assumed the student is expected to extrapolate these standard rules from the diagram.
- **Cofunction Reversal:** In Example 2d (Page 4), the expression includes $\cos(y - \frac{\pi}{2})$. The cofunction identity given on page 2 is specifically

$\cos(\frac{\pi}{2} - y) = \sin y$. To solve 2d, one must use the even identity $\cos(-\theta) = \cos(\theta)$ to flip the terms. I resolved this by assuming the student knows to apply the even-odd property to match the cofunction template.

- **Absence of an Answer Key:** The provided PDF is a worksheet full of problems with no solutions attached. All steps, solutions, and verifications provided in the study guide and practice sections are my own mathematical derivations based on the rules outlined in the text.